

BorrowDesk: Equity-Collateralized USDG Credit on Robinhood Chain

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Status: Live mainnet (Robinhood Chain 4663) · Arbitrum Open House Singapore Online Buildathon

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App: <https://borrowdesk.fun> · **Repo:** github.com/AmaanSayyad/BorrowDesk

Abstract

BorrowDesk is a lending market on **Robinhood Chain** where holders of official Stock Tokens (NVDA, AAPL, TSLA, SPY, and 25 more) deposit equity exposure as collateral and borrow **USDG (Global Dollar)** without selling.

The protocol is two verified contracts — **BorrowDeskMarket V2** and **BorrowDeskLens** — plus a credit-desk UI that shows liquidation math **before** you sign. This paper explains how it works in plain language, with the key formulas, diagrams, and references a builder would want to check.

1. Why this exists

Robinhood Chain already has the raw materials: Stock Tokens as ERC-20s with Chainlink price feeds [1–3], and USDG as the dollar rail [4,5]. Robinhood’s own docs even describe “deposit stocks → borrow USDG” as a thing you can build [6].

What was missing was a **desk** for that flow:

- one account across many stocks (not a separate vault per ticker)
- risk that survives equity weekends (feeds don’t update like crypto 24/7)
- a UI that shows liq price and buffer **before** the wallet popup

Morpho-style isolated markets are great for curators [7,8]. BorrowDesk is for the operator who wants one book, clear LTV bands, and foresight — not maximum TVL cosplay.

2. What we shipped (the “invention” in practice)

Four ideas work together. None of them needs a PhD; together they are what makes BorrowDesk feel different from a generic money-market fork.

Idea	Plain English	Onchain / product
Multi-collateral account	NVDA + AAPL share one debt line	One <code>Account</code> mapping + one borrow index
Soft-session oracle	Tolerate Friday's mark over the weekend; fail if the feed is truly dead	<code>maxOracleDelay = 4 days</code>
Foresight ticket	"If NVDA drops X%, you're liquidatable" before you sign	Desk math on Lens health + LTV
Atomic ticket	Deposit and borrow in one tx	<code>depositAndBorrow</code>

Health is shown as **0–10** on the desk (capped). Under the hood it's still the usual ratio: liquidation capacity ÷ debt.

3. Architecture

3.1 System overview

Figure 1 - BorrowDesk system architecture (Robinhood Chain 4663)

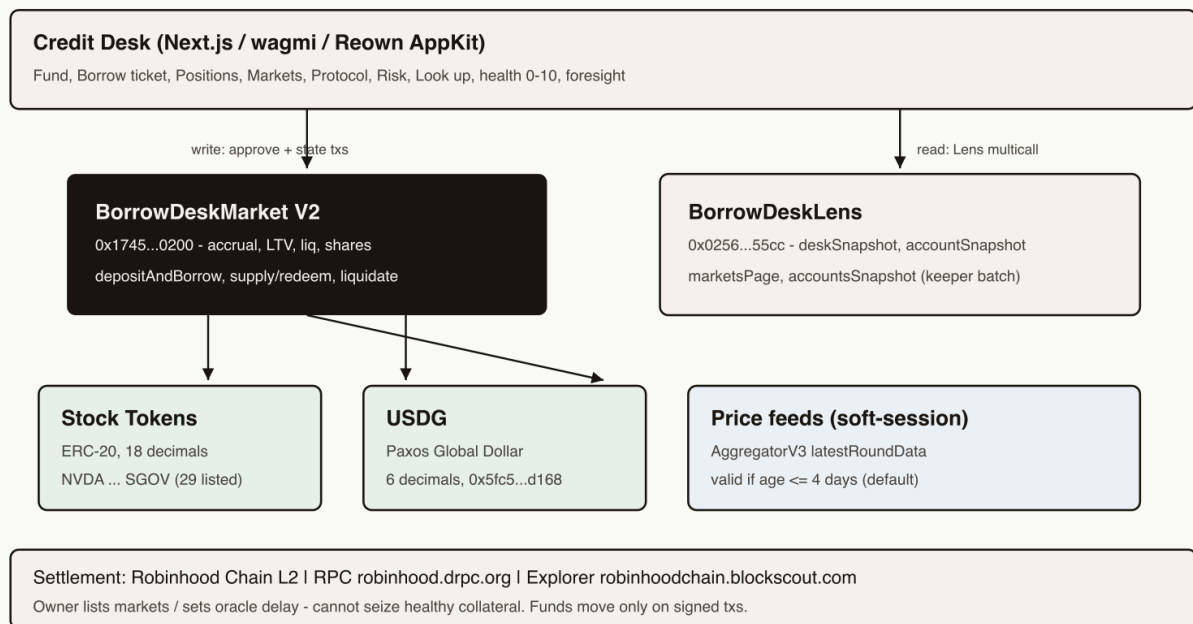


Figure 1. Desk writes to Market V2; reads batch through Lens. Stock Tokens + USDG settle on Robinhood Chain; price feeds gate every risk check.

3.2 Stack

Figure 2 - Protocol stack: product layer vs rails

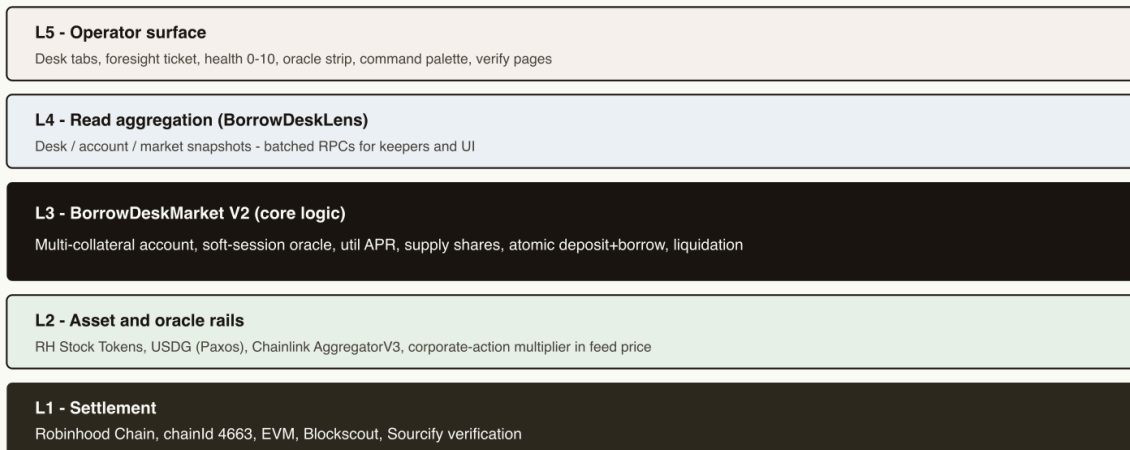
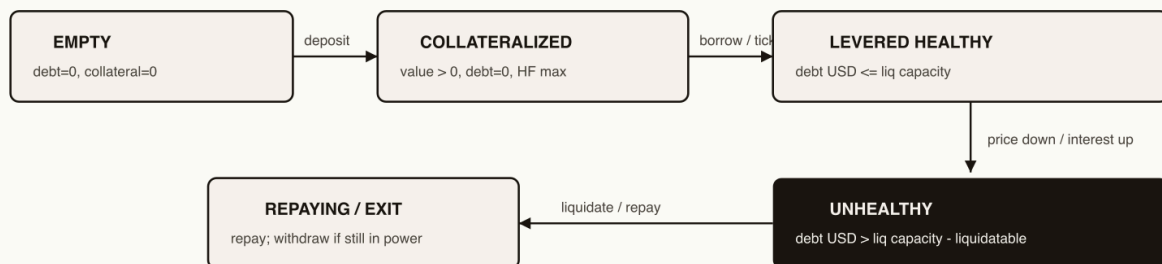


Figure 2. L1–L2 are commodity rails (chain, tokens, oracles). L3 is BorrowDesk’s market logic. L4–L5 are Lens + the desk UI.

3.3 Account lifecycle

Figure 3 - Account state machine and interest index



Global borrow index (Compound-style)

Index grows with time and utilization. User debt = principal x currentIndex / snapshotIndex.

Borrow APR ~ 2% + 8% x utilization (about 2% empty, up to ~10% fully used).

Every asset-moving call accrues first. Interest is absorbed into principal via the index - no separate interest bucket.

depositAndBorrow is one transaction: collateral in and USDG out with a single borrow-power check.

Withdraw while leveraged requires post-state debt ≤ borrow power (stricter than liquidation line).

Figure 3. Empty → deposit → borrow → healthy leverage. If prices fall or interest accrues enough, the account becomes liquidatable. Repay or get liquidated back toward safety. Withdrawals must leave you inside **borrow power** (stricter than liquidation threshold).

4. How the market works

4.1 Collateral and debt

You deposit listed Stock Tokens. You borrow USDG from the pool. Each listed asset has:

- **LTV** — max you can borrow against it
- **Liquidation threshold** — above this debt ratio, liquidators can act
- **Liquidation bonus** — extra collateral liquidators receive (capped at 20%)

Example bands on mainnet (bps → percent): mega-caps often ~60% LTV / 75% liq; higher-beta names tighter (e.g. TSLA ~50% / 65%). Full list: `deployments/robinhood.json`.

All values are converted to USD using the asset's AggregatorV3 feed. Internally the contract uses 18-decimal USD; USDG is 6 decimals, so:

$$\text{USD value of USDG} = \text{amount} \times 10^{12}$$

4.2 Health (simple version)

$$\text{Borrow power} = \sum_t (\text{value}_t \times \text{LTV}_t)$$

$$\text{Liq capacity} = \sum_t (\text{value}_t \times \text{liq threshold}_t)$$

- **Can borrow / withdraw?** debt ≤ borrow power
- **Healthy?** debt = 0, or debt ≤ liq capacity

Desk display:

$$\text{HF}_{0-10} = \min\left(10, \frac{\text{Liq capacity}}{\text{debt}} \times 10\right)$$

4.3 Interest

Interest uses a Compound-style global index [11]. Roughly:

$$\text{utilization } u = \frac{\text{total debt}}{\text{idle USDG} + \text{total debt}}$$

$$\text{borrow APR} \approx 2\% + 8\% \times u \quad \Rightarrow \quad \text{about 2\% empty pool, up to ~10\% fully used}$$

Every deposit / borrow / repay / liquidate accrues first. There is no separate “accrue” button in the product — it happens inside the market call.

4.4 Supplying USDG

Anyone can **supply** USDG and receive shares. Share value rises as borrowers pay interest:

$$\text{total assets} = \text{idle USDG} + \text{total debt}$$

Redeem needs enough **idle** cash. If almost everything is borrowed, suppliers may wait — normal money-market behavior.

5. Oracles and weekends

Figure 4 - Soft-session oracle validity window

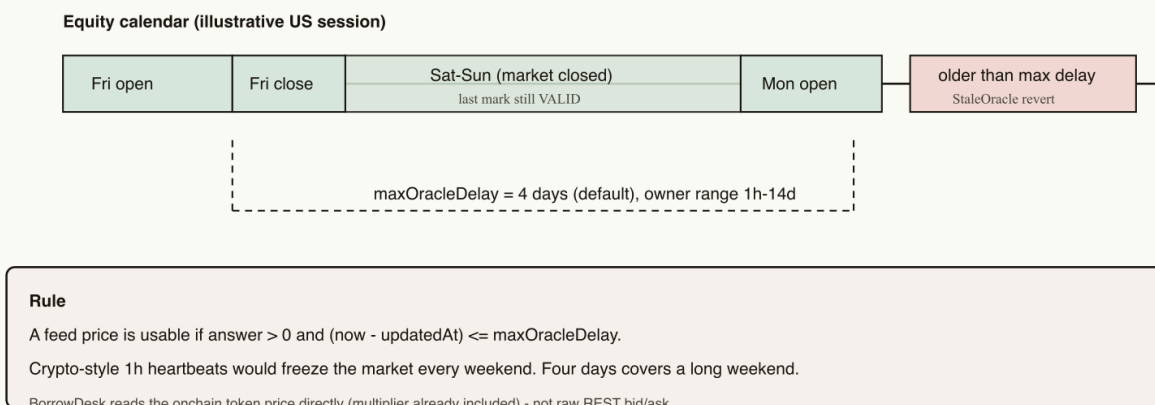
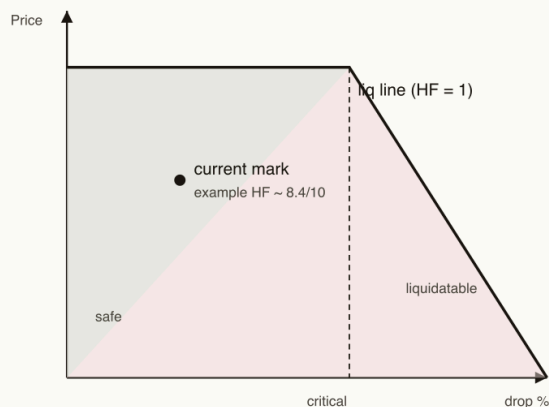


Figure 4. Equity markets close on weekends. A 1-hour crypto-style staleness check would freeze the market every Friday. BorrowDesk allows marks up to **4 days** old (owner can set 1 hour–14 days). It does **not** invent a Sunday price — it keeps using the last good mark until the feed updates or the window expires. Feeds return **token** price (share × corporate-action multiplier) [2,3]. Don't mix them with raw REST equity quotes without applying the multiplier [13].

6. Liquidation foresight

Figure 5 - Drop-to-liquidatable (foresight before you sign)



How the desk estimates it

Hold other prices fixed. Find the price of one asset where debt = liq capacity.

$$P^* = \frac{\text{debt USD} - \text{other liq value}}{\text{balance} \times \text{liq threshold}}$$

$$\text{drop \%} = \max(0, 1 - P^* / P_{\text{now}})$$

Borrow sizing uses LTV (borrow power).
Drop-to-liq uses liquidation threshold.

UI health: $\min(10, \text{liq/debt} \times 10)$

Onchain: healthy if debt $\leq$ liq capacity

Example: \$100 NVDA, \$50 debt, 75% liq
 $\Rightarrow \sim 33\%$ drop before liquidatable

Figure 5. Before you sign a borrow, the desk estimates how far a name can fall before you hit the liquidation line.

For a single stressed asset (other prices fixed), the critical price is:

$$P^* = \frac{\text{debt USD} - (\text{liq value of other collateral})}{\text{balance} \times \text{liq threshold}}$$

$$\text{drop \%} = \max\left(0, 1 - \frac{P^*}{P_{\text{now}}}\right)$$

Tiny example. 1 NVDA at 100, 7550 debt, no other collateral $\rightarrow$ liquidatable around \$66.67 $\rightarrow$ about a 33% drop. The ticket should say that up front.

Liquidators repay USDG on unhealthy accounts and seize collateral with a bonus. Partial repay is allowed (no close-factor in V2).

Atomic path for one-click tickets:

```
depositAndBorrow(token, depositAmount, borrowAmount)
→ pull stock → raise debt → check borrow power → push USDG
```

One signature, one health check — no gap between “I deposited” and “I borrowed” across two txs.

7. Contracts and desk

Piece	Address	Role
Market V2	<code>0x1745...0200</code>	Deposits, borrows, supply, liquidations · Sourcify
Lens	<code>0x0256...55cc</code>	Batched reads for desk + keepers · Sourcify
USDG	<code>0x5fc5...d168</code>	Borrow asset · 6 decimals [5]
Legacy V1	<code>0x2e4E...9E8d</code>	Drained / unused

User funds move only when the wallet signs `approve` + market calls. The owner can list markets and set oracle delay — **not** seize healthy collateral.

Desk tabs: Overview · Borrow · Fund · Positions · Markets · Protocol · Risk · Look up · Start. Fund covers get-stock, supply USDG, and batch deposit+borrow. Wallet via Reown AppKit [14].

8. Risks (honest)

1. **Weekend marks stick** — Friday price may not equal Monday open.
2. **No close factor yet** — liquidations can take a large bite.
3. **Supplier exits need idle cash** — high utilization can delay redeems.
4. **Owner key** — bad listings (rogue feed) are the main admin risk until ownership hardens.
5. **Small inventory today** — rates and liquidations behave differently at scale.
6. **Stock Tokens ≠ shares** — legal wrapper of the issuer applies [1].
7. **No formal audit yet** — Sourcify proves source matches bytecode; it does not prove economic safety.

Report security issues privately — don't post exploit PoCs against live funds.

9. Competitive landscape (short)

Venue	Style	BorrowDesk difference
Morpho on RH [7,8]	Isolated markets / vaults	We optimize for one multi-stock book + foresight desk
Acre [15]	Multi-collateral USDG	Closest peer; we push desk UX, 29 feeds, public Lens/V2 verify
Zona [16]	One equity at a time; 24/5 window	We keep multi-collateral + 4d soft oracle
Pledge [17]	Isolated per ticker	Shared account + atomic ticket
Ondo / xStocks [18,19]	Other chains	Not RH Stock Tokens / USDG on 4663

10. Economics

Today: borrowers pay ~2–10% APR by utilization; suppliers earn that interest via share value; liquidators earn the bonus. No protocol fee. No product token.

Later: planned 1% **borrow fee**, a **protocol token** for governance / incentives (USDG stays the borrow asset), and interest splits as the pool grows.

11. Formula card

```

utilization      = debt / (idle + debt)
borrow APR       ≈ 2% + 8% × utilization
debt(user)       = principal × borrowIndex / snapshot
borrow power     = Σ collateralUSD × LTV
liq capacity     = Σ collateralUSD × liqThreshold
healthy          = debt == 0 OR debt ≤ liq capacity
HF (0–10)        = min(10, liq capacity / debt × 10)
drop to liq      = max(0, 1 - P* / P_now)
oracle OK        = price > 0 AND age ≤ 4 days (default)

```

12. Deployment

Item	Value
Chain	Robinhood Chain mainnet (4663)
Market V2	0x17452DAB5976B770107c126fBC92aD746a3C0200
Lens	0x025694b88ddd0a6ffac740df4170b8ea590355cc
Owner	0x1881Dfd2b29536F054AA0b0A4966856290388Cc2
Listed markets	29 Stock Token / ETF feeds
App	https://borrowdesk.fun
Pitch	Chronicle
Demo	https://youtu.be/KuaS7UGdRvY
Hackathon	Arbitrum Open House Singapore [20]

13. Closing

BorrowDesk is small on purpose: one market contract you can read, a lens for the desk, and UI that refuses to hide liquidation risk. The useful idea is simple — **bring dollar credit to where the stocks already are**, keep the oracle honest about weekends, and show the knife edge before anyone leans on it.

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